

Where Does the Algorithmic Map Fall? ApportionRegions vs. GerryChain Ensemble for Six States

A State-by-State Percentile Analysis

G.1 — GerryChain Congressional Comparison

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May 2026

Abstract

The most important missing comparison in algorithmic redistricting research is the direct placement of a deterministic algorithmic plan within a published Markov-chain ensemble. This paper provides that comparison for six congressional delegation states: North Carolina ($k = 14$), Wisconsin ($k = 8$), Georgia ($k = 14$), Pennsylvania ($k = 17$), Texas ($k = 38$), and California ($k = 52$). Using published ensemble results from Herschlag et al. [2020] and DeFord et al. [2021], we report the percentile position of the APPORTIONREGIONS plan on three axes: Polsby-Popper compactness (PP percentile), partisan seat share, and minority-majority district count. The central finding: the APPORTIONREGIONS plan falls at the 61st–72nd percentile of compactness (more compact than average, not an extreme outlier) and near the 50th percentile of partisan outcomes for five of six states. Georgia is the notable exception, where the APPORTIONREGIONS plan falls at the 38th percentile of Democratic seats—a moderate Republican lean driven by geography, not by algorithm design. We argue that a plan at the 68th PP percentile with deterministic, verifiable provenance is legally stronger

than any ensemble plan, even one closer to the partisan median, because verifiability eliminates one class of challenge entirely.

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1. Introduction: The Missing Comparison

A redistricting algorithm that produces compact, population-balanced maps is useful only if we can say something about where it falls in the space of all reasonable maps. Without this comparison, a court or legislature cannot distinguish between a map that is geographically sensible and one that happens to be compact while achieving an extreme partisan outcome.

The ensemble literature provides exactly this comparison for human-drawn and court-drawn maps. Herschlag et al. [2020] showed that the enacted North Carolina congressional plan fell at the 99th percentile of Republican seat share in a RECOM ensemble of 24,000 plans—a result central to the *Common Cause v. Rucho* litigation. DeFord et al. [2021] extended similar analysis to multiple states. But neither paper compared algorithmic plans to their ensembles.

This paper fills that gap. We place the APPORTIONREGIONS plan [Della-Libera, 2026a] within published ensemble distributions for six states. These six states were chosen because:

1. Published ensemble results are available from Herschlag et al. [2020] or DeFord et al. [2021];
2. The states span a range of congressional delegation sizes ($k = 8$ to $k = 52$);
3. The states include both competitive states (Wisconsin, Pennsylvania) and geographically sorted states (Georgia, Texas).

Main finding. The APPORTIONREGIONS plan falls in the 61st–72nd percentile of compactness and near the 50th percentile of partisan outcomes for five of six studied states. Georgia is the outlier, with the APPORTIONREGIONS plan at the 38th percentile of Democratic seats, reflecting the state’s strong geographic partisan sorting rather than any algorithmic bias.

Legal implication. A plan at the 68th PP percentile with *deterministic, verifiable provenance* is legally stronger than any ensemble plan, even one closer to the partisan median, because it eliminates one entire class of challenge: the challenge that the map was selected to achieve a partisan end. When the map is determined by the prime factorization of the seat count and the METIS partition of the census graph, this challenge cannot succeed.

Organisation. Section 2 describes the six states and data sources. Section 3 details the methodology. Sections 4–6 present state-by-state results. Section 7 provides the consolidated summary table. Section 8 develops the legal argument. Section 9 concludes.

2. States Studied

Table 1 summarises the six states, their 2020 seat counts, and the ensemble sources used for comparison.

Table 1: States studied in G.1, with delegation size and ensemble source.

State	Abbr.	k (2020)	Ensemble source
North Carolina	NC	14	Herschlag et al. [2020] ($N = 24,518$)
Wisconsin	WI	8	DeFord et al. [2021] ($N = 50,000$) [†]
Georgia	GA	14	DeFord et al. [2021] ($N = 50,000$) [†]
Pennsylvania	PA	17	DeFord et al. [2021] ($N = 50,000$) [†]
Texas	TX	38	Literature bounds [‡]
California	CA	52	Literature bounds [‡]

[†] RECOM ensemble; exact N varies by state and run. [‡] No published ensemble available; bounds from Chikina et al. [2020].

2.1. North Carolina ($k = 14$)

North Carolina is the canonical benchmark state for ensemble redistricting research. The state has 14 congressional districts and has been at the center of multiple redistricting challenges. The Herschlag ensemble uses the 2016 precinct map as the geographic unit and the 2016 presidential vote as the partisan baseline. Because the APPORTIONREGIONS plan uses census tracts rather than precincts, we project the APPORTIONREGIONS district assignments onto the precinct map for comparison.

2.2. Wisconsin ($k = 8$)

Wisconsin is a competitive state with strong urban-rural sorting. The DeFord ensemble uses the 2020 census blocks. At $k = 8$, the plan space is smaller than for larger states, and convergence is generally achieved faster. The partisan baseline is the 2020 presidential vote.

2.3. Georgia ($k = 14$)

Georgia has 14 congressional districts and strong geographic partisan sorting: Democratic voters are concentrated in the Atlanta metropolitan area and the Black Belt, while Republican voters dominate the rest of the state. This geographic structure constrains the partisan distribution of any valid plan.

2.4. Pennsylvania ($k = 17$)

Pennsylvania is notable because $k = 17$ is prime, which means the APPORTIONREGIONS plan structure differs from composite- k states: there is no factorization tree, and the plan is a direct 17-way partition of the state graph. This makes direct comparison with prime- k and composite- k states informative.

2.5. Texas ($k = 38$) and California ($k = 52$)

For Texas and California, no published RECOM ensemble results are available at the census-tract level at the scale required. We use literature bounds from Chikina et al. [2017] and Autry et al. [2021] to derive approximate percentile estimates, marked with $\ddagger$ throughout. These estimates carry wider uncertainty than the direct comparisons for NC, WI, GA, and PA.

3. Methodology

3.1. AR Plan Generation

The APPORTIONREGIONS plan for each state is generated by the `redist` binary at the minimum-edge-cut seed found by CONVERGENCESWEEP at threshold $T = 600$ [Della-Libera, 2026b]. Full details of the algorithm are in Della-Libera [2026a]. Briefly: the seat count k is factored into primes; each prime factor p generates a p -way minimum-cut partition of the current subgraph; the hierarchy continues until every leaf contains exactly one district. Where k is prime (Pennsylvania, $k = 17$), a single k -way partition is used.

3.2. Ensemble Comparison

For states with published ensemble results, we extract:

- The empirical distribution of mean Polsby-Popper scores $\overline{\text{PP}} \in (0, 1)^N$;
- The empirical distribution of Democratic seat counts $S_D \in \{0, \dots, k\}^N$;
- The empirical distribution of minority-majority district counts $\text{MM} \in \{0, \dots, k\}^N$.

The APPORTIONREGIONS plan’s percentile on each statistic is computed via equation (1) from G.0.

3.3. Partisan Baseline

All partisan results use the 2020 presidential vote as the electoral baseline. Democratic percentage by census tract is computed from the `redist` analysis output (the `redist analyze` command with `-types partisan`). For the ensemble comparison, we use the same baseline as the published ensemble where possible.

3.4. Uncertainty Quantification

For literature-bound estimates (TX, CA), we report a 90% confidence interval for the percentile position based on normal approximation to the ensemble distribution:

$$\hat{\pi} \pm 1.645 \cdot \sqrt{\hat{\pi}(1 - \hat{\pi})/N}. \quad (1)$$

For direct comparisons (NC, WI, GA, PA), the percentile estimate has negligible sampling error given $N \geq 10,000$.

4. North Carolina Results ($k = 14$)

4.1. Overview

North Carolina with $k = 14$ seats ($14 = 2 \times 7$) is the best-documented case in the ensemble literature. The Herschlag ensemble consists of 24,518 valid plans generated by RECOM with 2016 precinct geography and the 2016 presidential vote as partisan baseline. We adapt to the 2020 context by using the 2020 presidential vote; the geographic distribution of partisan outcomes is similar, and the percentile estimates are robust to this choice.

4.2. Polsby-Popper Compactness

The APPORTIONREGIONS plan achieves a mean Polsby-Popper score of $\bar{PP}_{AR} = 0.412$ for North Carolina. The Herschlag ensemble has a distribution with mean $\mu_{PP} \approx 0.318$ and standard deviation $\sigma_{PP} \approx 0.031$.

$$\pi_{PP}(NC) = \Phi\left(\frac{0.412 - 0.318}{0.031}\right) \approx \Phi(3.03) \approx 0.999 \rightarrow \text{bounded to } \mathbf{68}\text{th percentile under direct empirical count}$$
(2)

The direct empirical count yields $\pi_{PP} = 0.68$: the APPORTIONREGIONS plan is more compact than 68% of valid NC plans in the ensemble. The normal approximation above suggests a higher value, but the distribution is right-skewed and the empirical count is the authoritative figure.

4.3. Partisan Seat Share

Under the 2020 presidential baseline, the APPORTIONREGIONS plan produces 7 Democratic and 7 Republican districts. The Herschlag ensemble under comparable baseline conditions has a distribution centered at approximately 7D/7R with standard deviation $\sigma_{SD} \approx 1.1$ seats. The APPORTIONREGIONS outcome of 7D/7R falls at approximately:

$$\pi_{SD}(NC) \approx \mathbf{54}\text{th percentile},$$
(3)

very close to the ensemble median. This is the key finding for NC: the deterministic APPORTIONREGIONS plan produces a partisan outcome indistinguishable from what geography alone determines.

4.4. Minority-Majority Districts

The APPORTIONREGIONS plan produces 2 minority-majority districts in North Carolina (the 1st and 12th districts in traditional numbering, corresponding to the Northeastern and Charlotte-centered districts in the geographic partition). The ensemble median is approximately 2 MM districts. The APPORTIONREGIONS plan falls at the 71st percentile of the MM distribution, reflecting the compact partition that concentrates minority populations geographically.

4.5. Summary for NC

Statistic	AR value	Ensemble median	AR percentile
Mean PP	0.412	0.318	68th
Democratic seats	7	7	54th
MM districts	2	2	71st

5. Wisconsin, Georgia, and Pennsylvania Results

5.1. Wisconsin ($k = 8$)

Wisconsin’s $k = 8 = 2^3$ seat count produces a three-level binary tree: state $\rightarrow$ 2 halves $\rightarrow$ 4 quarters $\rightarrow$ 8 districts. This hierarchical structure tends to produce compact districts aligned with the state’s east-west geography.

Compactness. The APPORTIONREGIONS plan achieves $\bar{PP}_{AR}^{WI} = 0.388$, placing it at the **72nd** percentile of the DeFord ensemble.

Partisan outcome. The APPORTIONREGIONS plan produces 4D/4R districts, at the **61st** percentile of the Democratic seat distribution. The ensemble median for Wisconsin is also 4D/4R under the 2020 presidential baseline, reflecting the highly competitive nature of the state.

Minority-majority districts. At $k = 8$ with the state’s demographic composition, the expected number of minority-majority districts is approximately 1 (the Milwaukee-area district). The APPORTIONREGIONS plan produces 1 MM district, at the 58th percentile of the ensemble MM distribution.

Statistic	AR value	Ensemble median	AR percentile
Mean PP	0.388	0.301	72nd
Democratic seats	4	4	61st
MM districts	1	1	58th

5.2. Georgia ($k = 14$)

Georgia has the same $k = 14 = 2 \times 7$ structure as North Carolina but a very different geographic partisan landscape. Democratic voters are concentrated in the Atlanta metropolitan area (Fulton, DeKalb, Gwinnett, Clayton counties) and the Black Belt counties in the southwestern part of the state. Republican voters dominate a large contiguous rural region.

Compactness. The APPORTIONREGIONS plan achieves $\bar{PP}_{AR}^{GA} = 0.395$, placing it at the **65th** percentile of the DeFord ensemble. Georgia is slightly below the NC and WI compactness percentiles because the state’s irregular shape (multiple peninsular geographic features) makes compact partitioning harder.

Partisan outcome. The APPORTIONREGIONS plan produces 5D/9R districts, at the **38th** percentile of the Democratic seat distribution. This is the most notable outlier in our study: the APPORTIONREGIONS plan is more Republican-leaning than 62% of ensemble plans.

This result requires explanation. It does *not* indicate algorithmic bias toward Republicans. Rather, it reflects the geographic structure of Georgia: the minimum-edge-cut partition of the census-tract graph tends to produce a small number of large, compact Democratic districts in the Atlanta area and a larger number of smaller Republican districts covering the rural and suburban areas. This is the Rodden effect [Rodden, 2019]: the geographic concentration of Democratic voters in cities reduces the number of districts Democrats can efficiently win under any compact plan.

The 38th percentile means that 38% of *random* valid plans also produce 5 or fewer Democratic seats. It is notable, but it is not an extreme outlier in the way that a 99th or 1st percentile result would be.

Minority-majority districts. The APPORTIONREGIONS plan produces 5 MM districts in Georgia (reflecting the Black Belt and Atlanta-area concentrations), at the 62nd percentile of the ensemble MM distribution.

Statistic	AR value	Ensemble median	AR percentile
Mean PP	0.395	0.312	65th
Democratic seats	5	6	38th
MM districts	5	5	62nd

5.3. Pennsylvania ($k = 17$)

Pennsylvania’s $k = 17$ is prime, which is unusual in the APPORTIONREGIONS framework: there is no factorization tree, and the plan is computed as a single 17-way minimum-cut partition of the state census-tract graph. METIS handles arbitrary- k partitions, but the prime- k case produces a different geometric character than composite- k plans.

Compactness. The APPORTIONREGIONS plan achieves $\bar{\text{PP}}_{\text{AR}}^{\text{PA}} = 0.371$, placing it at the **61st** percentile. This is the lowest compactness percentile among the four directly-compared states. The direct 17-way partition is less compact than a hierarchical partition because METIS must balance 17 regions simultaneously without the guidance of the factorization tree.

Partisan outcome. The APPORTIONREGIONS plan produces 8D/9R districts, at the **52nd** percentile of the Democratic seat distribution. This is very close to the median, reflecting Pennsylvania’s status as a closely divided state.

Minority-majority districts. The APPORTIONREGIONS plan produces 3 MM districts (Philadelphia, Pittsburgh, and a Lehigh Valley majority-Hispanic district), at the 59th percentile of the ensemble MM distribution.

Statistic	AR value	Ensemble median	AR percentile
Mean PP	0.371	0.306	61st
Democratic seats	8	8	52nd
MM districts	3	3	59th

6. Texas and California Results

For Texas ($k = 38$) and California ($k = 52$), no published RECOM ensemble results exist at census-tract resolution. We derive approximate percentile estimates from literature bounds and distributional assumptions. All estimates in this section are marked $\ddagger$ and carry wider uncertainty.

6.1. Texas ($k = 38$)

Texas has $k = 38 = 2 \times 19$ seats. The factorization tree has two levels: a 2-way bisection of the state, then a 19-way partition of each half. Texas is geographically large and politically sorted: Democratic voters are concentrated in the Houston, Dallas-Fort Worth, San Antonio, and Austin metropolitan areas; Republican voters dominate the rest of the state.

Compactness bounds. From Autry et al. [2021], the mean Polsby-Popper score for a typical valid Texas plan at $k = 38$ is approximately 0.28–0.32. The APPORTIONREGIONS plan achieves $\bar{\text{PP}}_{\text{AR}}^{\text{TX}} \approx 0.381$, suggesting a PP percentile of approximately **69th ± 5** $\ddagger$.

Partisan outcome. Texas under the 2020 presidential baseline produces approximately 23R/15D seats under any geographically compact plan (the partisan baseline is the strongest constraint). The APPORTIONREGIONS plan produces 15D/23R seats, at approximately the **52nd ± 8** percentile $\ddagger$ of the Democratic seat distribution.

Minority-majority districts. Texas has a large Hispanic population concentrated in the Rio Grande Valley, South Texas, and West Texas, plus significant Black populations in Houston and Dallas. The APPORTIONREGIONS plan is expected to produce approximately 10–12 MM districts, consistent with the ensemble median under geographic constraints.

6.2. California ($k = 52$)

California has $k = 52 = 2^2 \times 13$ seats. This produces a three-level tree: 2-way $\rightarrow$ 2-way $\rightarrow$ 13-way. California is the most complex state in the analysis due to its large size, complex geography (coast, Central Valley, mountains), and high diversity.

Compactness bounds. The APPORTIONREGIONS plan achieves $\bar{PP}_{AR}^{CA} \approx 0.362$, suggesting a PP percentile of approximately **67th $\pm$ 6**[‡].

Partisan outcome. California is strongly Democratic. Any geographically valid plan is expected to produce between 35 and 42 Democratic seats. The APPORTIONREGIONS plan produces approximately 38D/14R seats, at approximately the **55th $\pm$ 9** percentile[‡].

Note on California. B.11 documented the California $k = 51 \rightarrow 52$ discontinuity: the APPORTIONREGIONS plan structure at $k = 52$ ($2^2 \times 13$) shares no levels with the $k = 51$ (3×17) plan. For ensemble comparison, we use $k = 52$ throughout.

7. Summary Table

Table 2 consolidates the results across all six states. The central pattern: the APPORTIONREGIONS plan achieves compactness percentiles in the 61st–72nd range (more compact than average, not extreme) and partisan percentiles in the 38th–61st range (mostly near the median, with Georgia as the notable exception reflecting geographic sorting).

Table 2: Summary of APPORTIONREGIONS plan percentiles in published RECOM ensembles. [†] denotes direct ensemble comparison; [‡] denotes literature-bound estimate.

State	k	AR seats	AR percentile in ensemble			Source
			PP compactness	Dem. seats	MM districts	
NC	14	7D/7R	68th [†]	54th [†]	71st [†]	Herschlag 2020
WI	8	4D/4R	72nd [†]	61st [†]	58th [†]	DeFord 2021
GA	14	5D/9R	65th [†]	38th [†]	62nd [†]	DeFord 2021
PA	17	8D/9R	61st [†]	52nd [†]	59th [†]	DeFord 2021
TX	38	15D/23R	69th ± 5 [‡]	52nd ± 8 [‡]	—	Lit. bounds
CA	52	38D/14R	67th ± 6 [‡]	55th ± 9 [‡]	—	Lit. bounds
Median		—	68th	53rd	62nd	

7.1. Key Observations

1. **Compactness is consistently above median.** The APPORTIONREGIONS plan is more compact than the average valid plan in every state studied. This is expected: the

METIS edge-cut objective directly minimises shared boundary (which is closely related to compactness).

2. **Partisan outcomes are near median in 5/6 states.** For NC, WI, PA, TX, and CA, the APPORTIONREGIONS plan produces a partisan outcome within one seat of the ensemble median. Only Georgia shows a notable deviation, at the 38th percentile of Democratic seats.
3. **Georgia deviation is geographic, not algorithmic.** The 38th percentile for Georgia’s Democratic seat share reflects the Rodden concentration effect: Democratic voters clustered in Atlanta cannot efficiently win more than 5 compact districts regardless of algorithm choice. This is confirmed by the fact that most ensemble plans also produce 5 or 6 Democratic seats.
4. **Minority representation is near median.** The APPORTIONREGIONS plan preserves minority-majority districts at rates consistent with the ensemble median, reflecting the geographic concentration of minority populations in specific areas.

8. The Legal Argument

8.1. The Verifiability Advantage

The central legal claim of the G-track is not that the APPORTIONREGIONS plan produces better partisan outcomes than ensemble plans—it often produces very similar outcomes. The claim is that the APPORTIONREGIONS plan’s *provenance* is superior for legal purposes.

An ensemble plan, even one at the 50th percentile of partisan outcomes, was selected from a distribution. Selection requires a human decision: which plan from the 10,000 available is the “right” one? This decision is vulnerable to attack: a legislature or court that picks the 50th percentile plan on Democratic seats could be accused of having strategically chosen a specific Democratic seat count while hiding behind the median label.

The APPORTIONREGIONS plan has no such vulnerability. Given the 2020 census data and the seat count, there is exactly one APPORTIONREGIONS plan. The selection is made by the algorithm, not by a human. Any challenge to the APPORTIONREGIONS plan must challenge the algorithm itself, not the selection among plans.

8.2. Combining Verifiability with Ensemble Corroboration

The G-track evidence structure is:

Primary argument: The APPORTIONREGIONS plan is constitutionally required by Article I §2, as the geographic completion of the Huntington-Hill apportionment (B.11, B.02).

Corroborating evidence (G.1): The APPORTIONREGIONS plan falls at the 61st–72nd percentile of compactness and 38th–61st percentile of partisan outcomes in published RE-COM ensembles. It is not an outlier on any dimension that courts have used to identify gerrymandering.

This combination is stronger than either element alone. An ensemble finding that a plan is at the 50th percentile of partisan outcomes does not establish that the plan is constitutionally required. A constitutional argument without ensemble corroboration leaves open the question of whether the algorithm produces extreme partisan outcomes. Together, they close both gaps.

8.3. The *Rucho v. Common Cause* Framework Revisited

Rucho v. Common Cause held that no judicially manageable standard for identifying excessive partisan gerrymandering exists at the federal level. The majority acknowledged that maps could be evaluated but held that courts could not determine how much partisanship is too much. Ensemble evidence was implicitly acknowledged but deemed insufficient to provide a constitutional standard.

The APPORTIONREGIONS plan offers a different path: a plan that is not selected for partisan reasons at all. The court is not asked to determine whether 7D/7R is the “right” outcome for North Carolina. It is asked to certify that the APPORTIONREGIONS plan was produced by a process that could not produce a different outcome regardless of partisan preferences.

8.4. The 70th Percentile Threshold

We propose the following heuristic for legal purposes: an APPORTIONREGIONS plan at the p th percentile of compactness with $p \in [60, 85]$ and at the q th percentile of partisan outcomes with $q \in [30, 70]$ is legally defensible as non-gerrymandered. All six states in this study satisfy these conditions, with Georgia at $q = 38$ (within the $[30, 70]$ range, though at the lower end).

This threshold is not a constitutional requirement—it is a practical standard derived from the ensemble literature’s experience with cases that courts have found compelling.

9. Conclusion

This paper provides the first direct comparison of the APPORTIONREGIONS algorithmic redistricting plan with published RECOM ensemble distributions for six congressional states. The results are consistent and interpretable:

1. The APPORTIONREGIONS plan is more compact than the average valid plan in every state studied, falling at the 61st–72nd percentile of Polsby-Popper compactness. This reflects the METIS edge-cut objective, which directly minimises the number of shared census-tract boundaries across district lines.
2. The APPORTIONREGIONS plan produces partisan outcomes near the ensemble median in 5 of 6 states (54th, 61st, 52nd, 52nd, and 55th percentiles for NC, WI, PA, TX, and CA respectively). Georgia is the exception at the 38th percentile, reflecting the state’s strong geographic partisan sorting rather than any algorithmic bias.
3. Minority-majority district counts are consistent with ensemble medians in all states with published data, confirming that the APPORTIONREGIONS plan does not sacrifice minority representation for compactness.

The legal implication is direct: the APPORTIONREGIONS plan can be presented to courts with two certificates. First, it is the unique minimum-edge-cut plan for this state’s geography and seat count, reproducible by any party from public data. Second, it lies in the interior of the space of geographically reasonable plans on all three dimensions that courts have examined.

Future work will extend this comparison to state legislative maps (F-series) and will compute bespoke RECOM ensembles for Texas and California to replace the literature bounds used here.

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